

Control work 4, and work on the mistakes

The congruence of triangles in one paper, and the geometry chapter closed.

LESSON 56–57

2 × 40 MIN

MATEMATIKA 7, NAZORAT ISHI 4

S8 5 REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The map	5 min

LEARNING OBJECTIVES

- Choose the right criterion from the given data.
- Write a proof in statement-and-reason form under time.
- Use the properties of isosceles triangles and the triangle inequality.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 118–151
Cambridge	Stage 8, pp. 48–56
Workbook	Control paper M4

01

Explanation

The paper — 25 marks, 40 minutes

Q	TASK	MARKS	FROM
1	Can a triangle have sides 5, 8, 14? And 6, 9, 13?	4	L46
2	An isosceles triangle has apex 44°: find the base angles	4	L49–50
3	$AB = AD$, $\angle BAC = \angle DAC$: prove $\triangle ABC \equiv \triangle ADC$	6	L47–48
4	Name the criterion for: three sides; a side and two angles; two sides and the angle between	3	L47–54
5	In a rhombus, prove that a diagonal bisects two angles	5	L53–54
6	Describe the construction of an angle bisector	3	L55

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for checking the longest side only; Q3 two for the common side and the named criterion; Q5 two for stating that all four sides of a rhombus are equal; Q6 one for “leave the arcs”.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
only two sides checked	$5 + 14 > 8$ ✓	check the longest: $5 + 8 < 14$
apex used as a base angle	$44^\circ, 44^\circ, 92^\circ$	$44^\circ, 68^\circ, 68^\circ$
common side omitted	two facts only	$AC = AC$
criterion not named	“so they are congruent”	“by the first criterion”
rhombus sides assumed unequal	“given $AB = AD$ ”	state that all four are equal
reasons missing	a list of true statements	a reason beside each
arcs rubbed out	a clean bisector	leave the construction visible

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

Chapter IV as one map

BLOCK	THE SENTENCE
triangles	classified by sides and by angles, independently
the triangle inequality	the longest side is less than the sum of the other two
the first criterion	two sides and the angle between them
isosceles triangles	equal sides $\iff$ equal base angles
the second criterion	a side and two angles
the third criterion	three sides — and rigidity
constructions	compasses and a straightedge, arcs left visible

LOOKING FORWARD

The rest of Quarter II is algebra: expressions, powers, monomials and polynomials. Geometry returns in Quarter III with parallel lines and the angle sum of a triangle.

02 Worked examples

EXAMPLE 1 Model answer, Q1: sides 5, 8, 14 and 6, 9, 13.

- ① Longest of the first is 14.
- ② $5 + 8 = 13 < 14$ — no triangle.
- ③ Longest of the second is 13.
- ④ $6 + 9 = 15 > 13$ — yes.

ANSWER

No; yes

EXAMPLE 2 Model answer, Q3: $AB = AD$, $\angle BAC = \angle DAC$.

- ① $AB = AD$ — given.
- ② $\angle BAC = \angle DAC$ — given, and it is the included angle.
- ③ $AC = AC$ — common side.
The mark most often missed.
- ④ $\triangle ABC \equiv \triangle ADC$ by the first criterion.

ANSWER

Proved by SAS

EXAMPLE 3 Model answer, Q5: a diagonal of a rhombus bisects two angles.

- ① All four sides of a rhombus are equal: $AB = AD$ and $CB = CD$.
- ② AC is common.
- ③ $\triangle ABC \equiv \triangle ADC$ by the third criterion.
- ④ CPCT gives $\angle BAC = \angle DAC$ and $\angle BCA = \angle DCA$.

ANSWER

Proved by SSS

03 Interactive model

Return Q3 with the reasons column blank and have the class fill it in; they discover that the reasons carry most of the marks.

Chapter IV in twelve questions

A triangle with two equal sides is:

scalene

isosceles

equilateral only

right

Can 2, 3, 6 be a triangle?

yes

no

sometimes

only if right

Congruent triangles have how many equal pairs?

3

4

6

2

The first criterion needs:

three sides

two sides and the included angle

a side and two angles

three angles

Base angles of an isosceles triangle are:

unequal

equal

right

obtuse

Apex 40° gives base angles:

40°

70°

100°

140°

The second criterion needs:

three sides

a side and two angles

three angles

two sides

The third criterion needs:

three sides

three angles

two sides

a side and an angle

A triangular framework is:

flexible

rigid

unstable

impossible

A common side in a figure gives:

nothing

an equality

an angle

a contradiction

Three equal angles give:

congruence

similarity only

nothing

rigidity

Construction arcs should be:

rubbed out

left visible

in pen

omitted

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Congruence	Tenglik	Равенство
Criterion	Alomat	Признак
Isosceles	Teng yonli	Равнобедренный
Triangle inequality	Uchburchak tengsizligi	Неравенство треугольника
Common side	Umumiy tomon	Общая сторона
Reason	Asos	Обоснование
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

For Q1 it is enough to check:

all three sums

the longest side

the shortest side

the perimeter

Q2 needs the angles to add to:

90°

180°

270°

360°

The missing fact in Q3 is:

an angle

the common side

a third side

nothing

Q5 uses:

SAS

ASA

SSS

none

Q6 must mention:

a protractor

leaving the arcs

measuring

a ruler

Work on the mistakes means:

fix the wrong line

rewrite the solution

copy the answer

skip it

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Can 5, 8, 14 be a triangle?

ANSWER No

2. Can 6, 9, 13 be a triangle?

ANSWER Yes

3. Apex 44° : base angles

ANSWER 68°

4. Three sides: which criterion?

ANSWER The third

5. A side and two angles: which?

ANSWER The second

6. Two sides and the included angle: which?

ANSWER The first

7. Three angles: which?

ANSWER None

Medium · the standard question

7 PROBLEMS

1. $AB = AD$, $\angle BAC = \angle DAC$: the third fact

ANSWER AC common

2. That proof uses

ANSWER The first criterion

3. A rhombus diagonal proof uses

ANSWER The third criterion

4. Base angle 68° : apex

ANSWER 44°

5. An isosceles triangle with perimeter 28 and base 8

ANSWER Sides 10, 10, 8

6. Sides 7, 7, 7: angles

ANSWER 60° each

7. Which framework is rigid?

ANSWER The triangle

Hard · stretch and reason

7 PROBLEMS

1. Range of c with $a = 6$, $b = 11$

ANSWER $5 < c < 17$

2. Integer values of c there

ANSWER 6 to 16

3. An isosceles triangle with an exterior base angle of 115° : the apex

ANSWER 50°

4. Prove the base angles equal, using the bisector from the apex

ANSWER SAS then CPCT

5. Prove the diagonals of a rhombus are perpendicular

ANSWER SSS then equal adjacent angles

6. Two triangles with $AB = DE$, $\angle A = \angle D$, $\angle B = \angle E$

ANSWER Congruent by ASA

7. Two triangles with $AB = DE$, $BC = EF$, $\angle A = \angle D$

ANSWER Not necessarily congruent

07 Homework

Homework — 5 tasks

Rewrite in full every proof that lost a mark, with a reason beside every statement.

- Can a triangle have sides 4, 7, 12? And 5, 7, 11?
- An isosceles triangle has apex angle 52° . Find the base angles.
- Prove that a diagonal of a rhombus bisects two of its angles.
- Name the criterion that applies to each of the three standard sets of data.
- Describe the construction of the bisector of an angle.